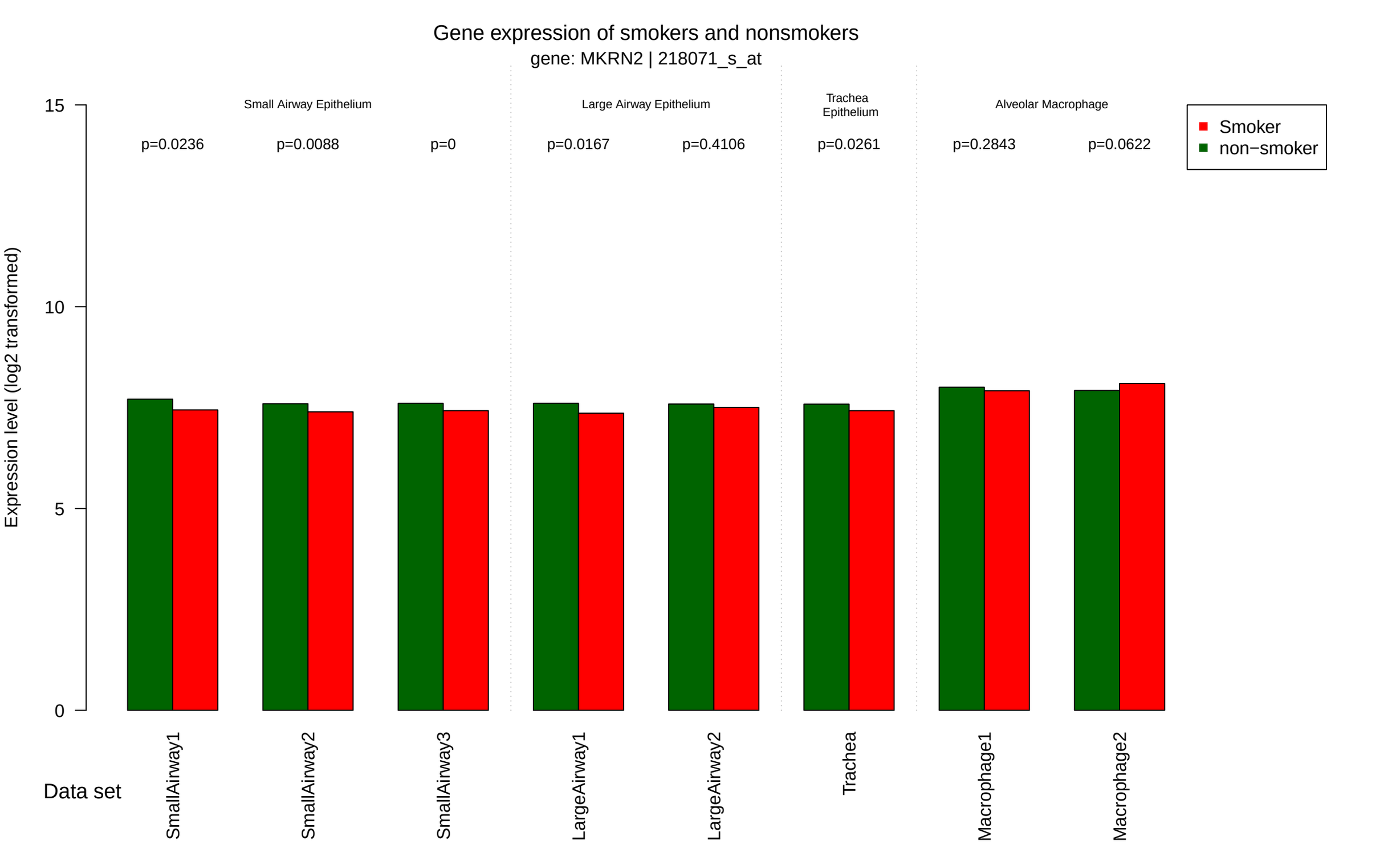
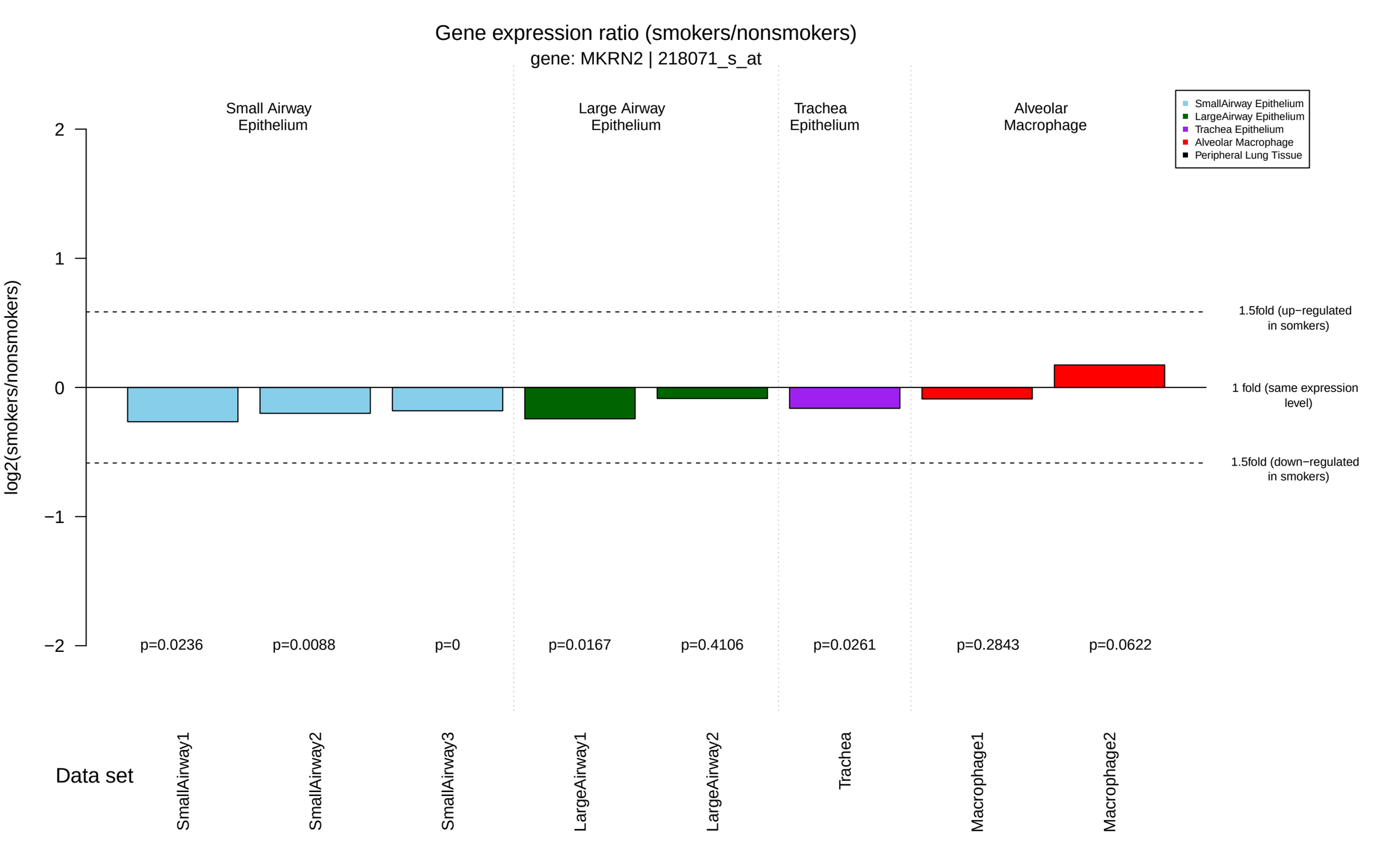
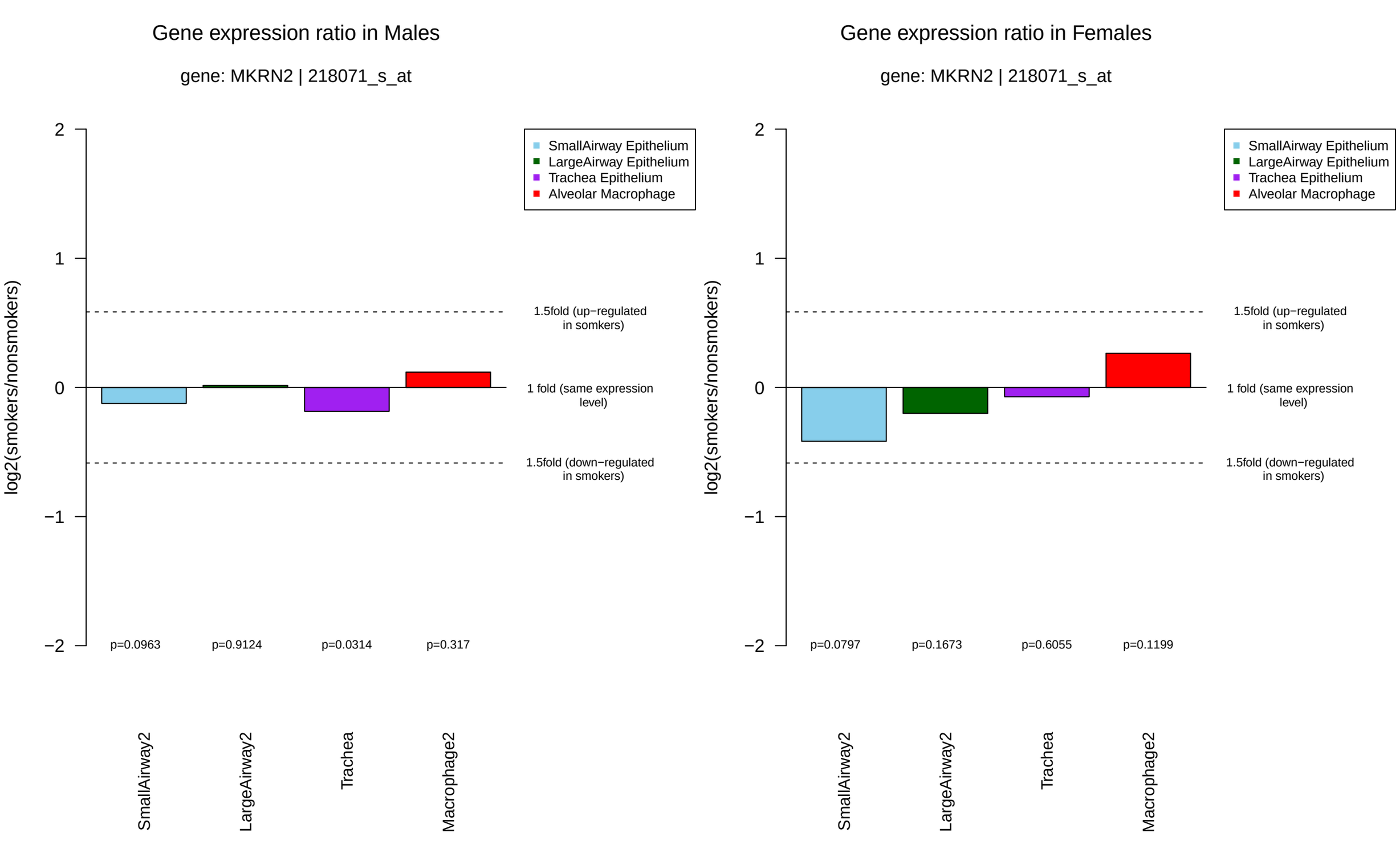
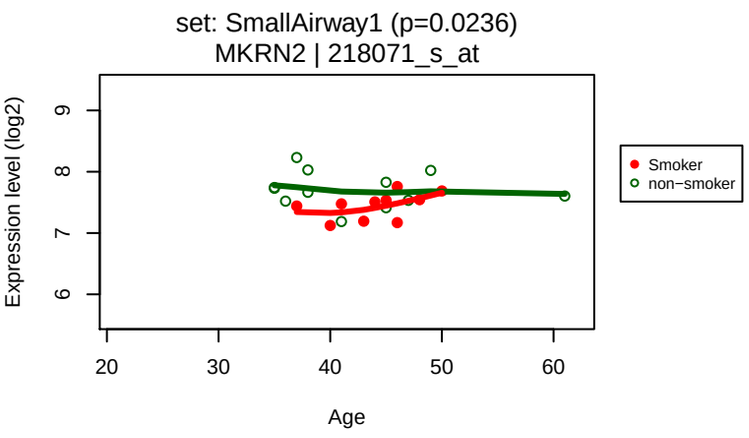
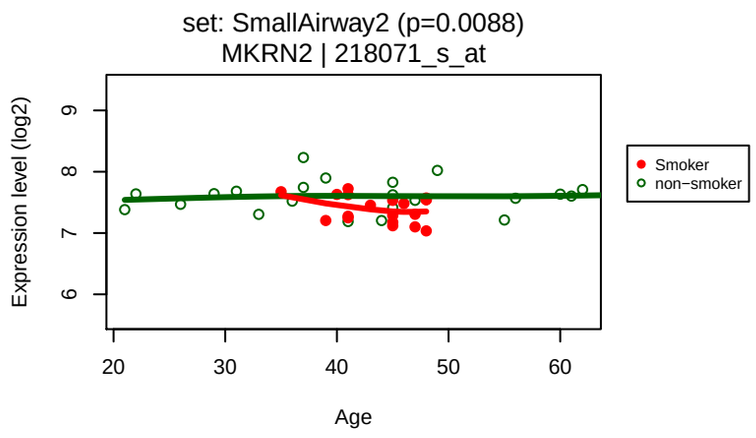
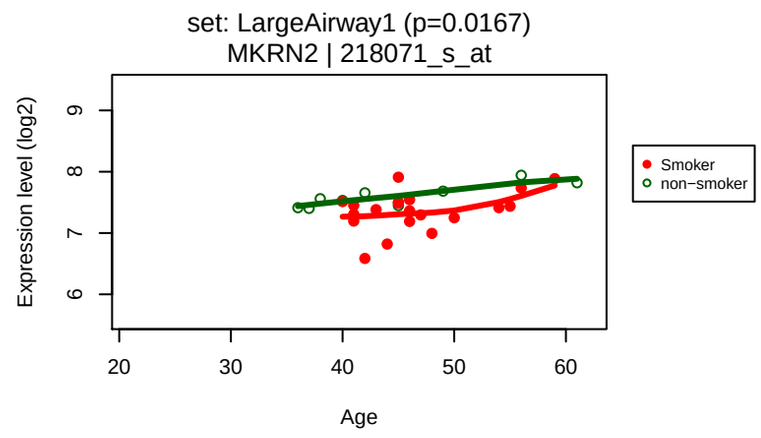

Gene expression of smokers and nonsmokers
gene: MKRN2 | 218071_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0236
p=0.0088
p=0
p=0.0167
p=0.4106
p=0.0261
p=0.2843
p=0.0622
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

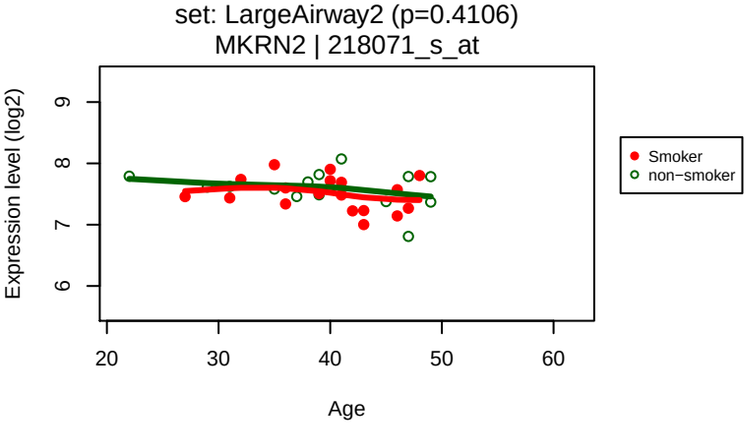

Gene expression ratio (smokers/nonsmokers)
gene: MKRN2 | 218071_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0236
p=0.0088
p=0
p=0.0167
p=0.4106
p=0.0261
p=0.2843
p=0.0622
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

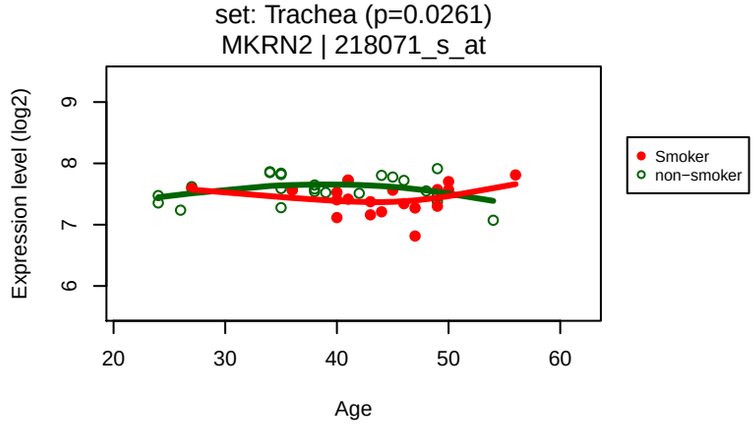

Gene expression ratio in Males
gene: MKRN2 | 218071_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0963
p=0.9124
p=0.0314
p=0.317
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MKRN2 | 218071_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0797
p=0.1673
p=0.6055
p=0.1199
SmallAirway2
LargeAirway2
Trachea
Macrophage2

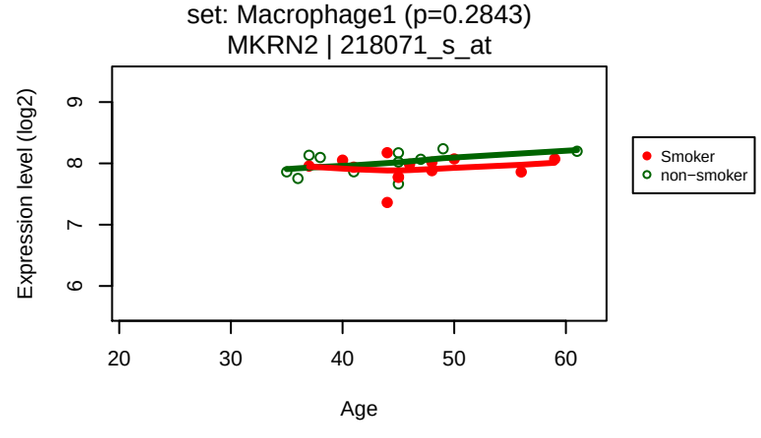

set: SmallAirway1 (p=0.0236)
MKRN2 | 218071_s_at
Expression level (log2)
Age
Smoker
non−smoker

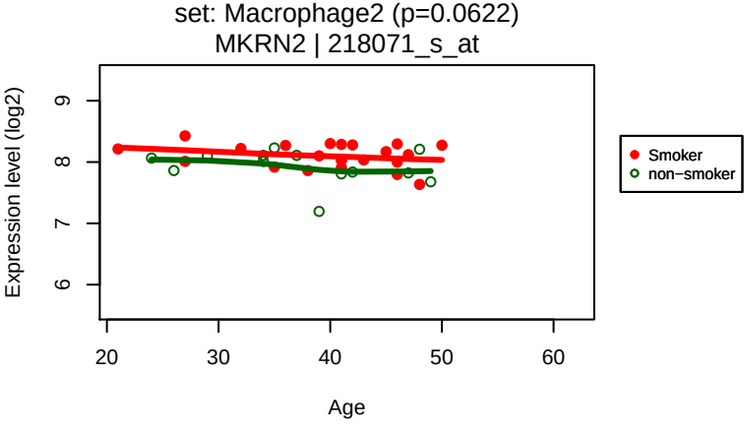

set: SmallAirway2 (p=0.0088)
MKRN2 | 218071_s_at
Expression level (log2)
Age
Smoker
non−smoker

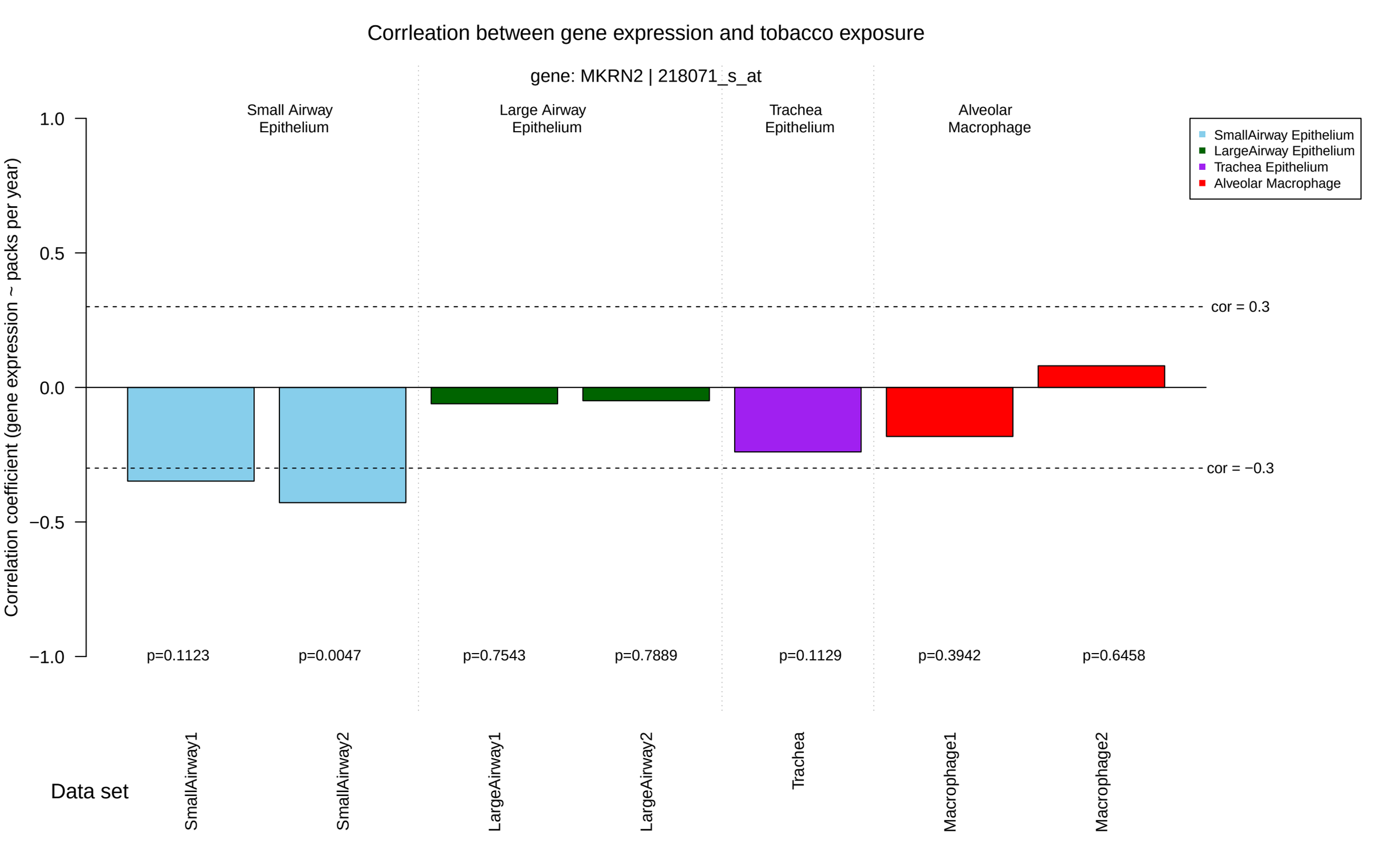

set: LargeAirway1 (p=0.0167)
MKRN2 | 218071_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4106)
MKRN2 | 218071_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0261)
MKRN2 | 218071_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2843)
MKRN2 | 218071_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0622)
MKRN2 | 218071_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MKRN2 | 218071_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1123
p=0.0047
p=0.7543
p=0.7889
p=0.1129
p=0.3942
p=0.6458
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2